Predator 8.004

Dense Closure Distillation

with a Preliminary Five-Agent Certificate-Regulated ALD Topology

Research brief for Brian Tenneson
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Core proposal

Train a general Predator policy once, then create a bounded, leakage-safe algebraic microtheory; exhaustively generate and verify a dense local closure; train on complete sets of legal proof actions and exact local proof depths; validate on withheld local theorems; freeze the model; and only then attempt the real theorem.

Status: Proposed architecture. Predator 8.003 remains a separate broad-training experiment and should be rerun unchanged after its search-control repair.

1. Executive summary

Predator 8.003 completed a large training run, but the local semigroup signal was sparse: only six theorem-level semigroup examples were available immediately before the withheld target sgrpcl. The resulting model may have learned broad proof-search regularities, yet it did not receive a dense mathematical education in the local algebraic domain. Separately, the first proof-search launch exposed a candidate-filtering defect: rough candidates were ranked and truncated before full unification, allowing inapplicable candidates to collapse the frontier. These are two distinct problems and require two distinct remedies.

- Repair 8.003 without retraining: unify first, rank only legal applications, correct exit-code classification, and add frontier diagnostics.
- Create 8.004 as a new experimental condition: broad pretraining plus dense local closure distillation and a readiness gate.
- Keep the final theorem and its proof strictly excluded from dense generation, expert iteration, and model selection.

Scientific separation

A repaired 8.003 tests broad historical transfer. Predator 8.004 tests whether broad transfer plus a dense local curriculum improves proof search. Mixing them would destroy the causal comparison.

2. Dense Closure Distillation

Dense Closure Distillation (DCD) converts a small, carefully bounded formal neighborhood into a large set of verified facts, proof states, legal actions, and exact local distances. It is a formal curriculum rather than a larger chronological window.

general set.mm pretraining
+

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bounded algebraic microtheory
    -> exhaustive verified closure
    -> policy + value training
    -> withheld local warm-up theorems
    -> freeze
    -> real withheld theorem
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2.1 Construct the local microtheory

For a semigroup target, the microtheory should include only chronologically legal pre-target material concerning extensible structures, base sets, binary operations, magmas, closure, associativity, identities, submagmas, homomorphisms, and the small logical toolkit needed to combine these facts. Selection should use dependency proximity, normalized syntax, symbol families, and co-citation, rather than declaration proximity alone.

2.2 Generate a dense verified closure

Within fixed bounds on expression size, inference depth, variable vocabulary, and total facts, a symbolic generator enumerates consequences. Every accepted fact carries a machine-checkable certificate. Duplicates, alpha-equivalent variants, and target-like leakage are removed. The corpus stores not only a successful proof, but the complete local decision context at each state.

Object	Stored information
Generated fact	Formula, assumptions, certificate, closure depth, source rules
Proof state	Open goals, active hypotheses, target, history, state hash
Legal application	Assertion, substitution, generated obligations, cost features
Alternative route	Other legal actions and alternate verified derivations
Failure information	Legal actions that led to dead or expensive states within the bounded microtheory

2.3 Train on the decision Predator actually makes

The symbolic environment first computes the genuinely unifying legal actions. The learning problem then ranks successful actions above other legal actions from the same goal. Pairwise same-goal ranking directly matches the ordering consumed by search and prevents goal-constant features from appearing informative.

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rough candidates -> full unification -> legal applications -> learned ranking -> controlled truncation
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2.4 Add a value or depth model

Because bounded closure provides exact local derivation depth, a second model can estimate remaining proof distance or probability of completion within budget. The policy chooses plausible actions; the value model discourages branches that are locally attractive but globally unproductive. Both remain advisory: only symbolically verified actions create successor states.

3. Efficient curriculum and readiness protocol

3.1 One expensive general model, many cheap local adaptations

The broad set.mm model should be trained once and cached. A target-family adaptation then trains for a short period on a balanced mixture of general examples, real local proofs, generated local closure facts, and hard states discovered during warm-up search. This is more efficient than replaying the entire database for every theorem.

Curriculum source	Purpose
General replay	Prevents catastrophic forgetting of broad logical and structural patterns.
Real local proofs	Anchors the curriculum to actual set.mm proof style.
Generated closure facts	Creates density and exact local depth labels.
Hard legal alternatives	Teaches distinctions among genuinely applicable actions.
Cross-agent first uses	Rewards lemmas that unlock another search process, not merely lemmas appearing in a final proof.

3.2 Expert iteration without target leakage

Predator attempts withheld development theorems from the same domain. Newly discovered verified proofs, useful first-use lemmas, and informative failed states are added to the curriculum for a fixed number of rounds. The real benchmark theorem remains excluded throughout. Any experiment that adapts on the real target must be labeled separately as online adaptation.

3.3 Readiness gate

The real theorem is attempted only after preregistered criteria are met:

- A strong solve rate on held-out local theorems under fixed budgets.
- Stable performance on symbol-renamed and meaning-preserving perturbations.
- Healthy frontier growth and no candidate-collapse events.
- Low invalid-proposal rate because legality is checked before ranking.
- Improvement over a compute-matched unguided baseline in expansions and wall time.
- No evidence that the model depends on target labels, close aliases, or downstream routes.

3.4 Proposed experiment

Condition	Training	Purpose
8.003 repaired	Existing broad historical model, unchanged	Measure broad transfer after fixing only the search defect
8.004 DCD	Same broad model plus dense local curriculum	Measure the causal effect of local closure distillation
Unguided	No learned score	Compute-matched symbolic baseline

4. Preliminary five-agent ALD topology

Brian proposes D = (R, N, C, I, V), with roles for proving c, proving not-c, finding inconsistency, certifying independence, and verifying certificates while regulating the shared lemma pool and traffic. This is a strong architecture. It combines a status-oriented ALD with a blackboard-style cooperative prover and a skeptical certificate checker.

4.1 A precise definition of p(c)

The notation p(c) can be retained as a defined predicate, but its certificate semantics should be explicit. For theory T, prover p, conjecture c, certificate pi, and checker V:

p(c) iff there exists pi such that Emit(p, c, pi) and V_T(pi, c) = ACCEPT.

This distinguishes producing a claim from proving it. A human prover may also act as the checker, so the logical definition permits a human certificate verifier to be the same person. Independence of the checker is an additional trust and experimental-design property, not part of the bare meaning of proof.

4.2 The five roles

Agent	Function
R - prover	Searches for a certificate of T |- c.
N - negator	Searches for a certificate of T |- not-c.
C - contradictor	Searches for T |- bottom, or for a target-local pair of accepted certificates for c and not-c.
I - independentator	Searches in a declared metatheory for accepted certificates that T does not prove c and T does not prove not-c.
V - verifier/regulator	Checks certificates, records assumptions and provenance, publishes compatible lemmas, rejects invalid entries, and schedules or throttles traffic.

4.3 Recommended refinement: split V internally

V should be one named role at the ALD level but two internal components in the implementation:

- V-kernel: a small deterministic checker that decides whether a proposed certificate is valid under its recorded assumptions.
- V-regulator: a heuristic scheduler and lemma-pool governor that assigns resources, deduplicates proposals, controls publication, and prevents traffic storms.

The regulator may be complex, learned, or buggy without threatening soundness. The kernel must remain small and trusted. Conflating them would enlarge the trusted computing base and make a scheduling fault look like a logical failure.

4.4 Evidence ledger and global status

D should not let agents directly announce final truth statuses. They submit evidence objects to V. V maintains a monotone ledger, and a separate status-extraction function computes the strongest licensed outcome.

Accepted evidence	Global status licensed
R certificate only	PROVED
N certificate only	REFUTED
R and N certificates, or a bottom certificate	INCONSISTENT
I certificates for both nonderivability directions, with compatible assumptions	INDEPENDENT
No conclusive evidence	UNKNOWN

Local process outcomes such as RUNNING, STOPPED, BUDGET_EXHAUSTED, FRONTIER_EMPTY, or FAULT belong to the audit log. They are not logical statuses. This distinction prevents a timeout from masquerading as independence and a search-control defect from masquerading as mathematical failure.

4.5 Prior art and likely contribution

Several ingredients are established: LCF-style systems use a small trusted proof kernel; proof-certificate frameworks separate an untrusted search procedure from a trusted checker; Leo-III and LeoPARD use multiagent blackboard architectures with shared proof state and scheduling; and the TPTP SZS ontology distinguishes success, unknown, stopped, and semantic statuses such as NoConsequence. The likely research contribution is therefore not any one ingredient, but the exact synthesis: role-specialized positive, negative, contradiction, and nonderivability agents; a certificate-indexed evidence ledger; and a universally accessible provenance-aware lemma pool regulated by a skeptical verifier.

Provisional originality judgment

The topology has recognizable ancestors, but I do not know of an established system identical to D = (R, N, C, I, V) with V simultaneously enforcing certificate validity, assumption compatibility, lemma publication, and traffic regulation. This is a plausible candidate-original synthesis, not a proven novelty claim.

5. Open design questions

- What exact local outcomes may R, N, C, and I report before V converts them into ledger entries?
- Does C search only for target-local inconsistency (c and not-c), or for arbitrary inconsistency of T?
- What metatheory certifies I, and how are its assumptions represented in the ledger?
- May V publish partially useful lemmas before a complete certificate is found?
- How are contradictory or assumption-incompatible lemma proposals quarantined?
- Which traffic policy preserves fairness so that a regulator cannot permanently starve a necessary low-ranked branch?
- Should D have one shared learned policy or role-specific policies with a solver selector?

6. Immediate implementation sequence

1. Repair and rerun Predator 8.003 with its saved model.
2. Build a leakage auditor and local-theory selector for semigroup-adjacent material.
3. Implement bounded verified closure generation and exact state/action logging.
4. Train a same-goal pairwise policy and a small value model.
5. Run withheld local warm-up theorems and symbol-renaming controls.
6. Freeze Predator 8.004 and compare it with repaired 8.003 and unguided search.

7. Implement D as an evidence-ledger shell around the Predator proposal agents, keeping V-kernel separate from V-regulator.

References and research basis

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Note on scope: This brief records a proposed research architecture and a preliminary analysis of the five-agent topology. The user had begun to specify R outcomes when the brief was prepared; those outcomes should be appended rather than silently inferred once supplied.